LearnMate- AI assistant for smarter learning and study support

Topic : AI – STUDY BUDDY API

Team number:05

Team head:PREETHI.S

81F8F7383E3A301B2414B94EDCC200D3

Team members :SIBIKA.R.S

2D50ECE8EBC652E9EC94B1BCE70E695F

Team members :SRUTHI.C

300D3172FE2F13897CF4652909151055

Team members :YOGASRI.M

07DF70DD2210435A4834A727229FC04E

Description

AIStudyBuddy is anAI-poweredlearningassistanceplatformdevelopedtosimplifythestudyprocess for students byleveraging modernwebtechnologiesandGenerative AI. ThebackendisdevelopedusingNode.jsandExpress.js,following aRESTful API architecture that ensures scalability, maintainability, and secure communication between the frontendandbackendservices. Passwordsaresecurelyencryptedusingbcryptjs,preventingunauthorizedaccesstousercredentials. One of the key highlights of AI StudyBuddy is its integration with the Google Gemini AI API, which enablestheapplicationtoperformadvancednaturallanguageprocessing tasks.

Scenario-Based Case Study

Background

Rahul is a second-year engineering student preparing for multiple semester examinations while simultaneously learning new technical skills for internships and placement opportunities. Due to the large volume of content and limited preparation time, Rahul often struggles to identify important concepts, create revision notes, and evaluatehisunderstandingbeforeexaminations.

Problem

Rahulencountersseveralchallengesduringhispreparation:

- Difficultyunderstandinglengthystudymaterialswithinlimitedtime.
- Manualpreparationofnotesconsumesasignificantamountofstudytime.
- Creatingflashcardsandrevisionquestionsisrepetitiveandlabor-intensive.
- Lackofpersonalizedstudyplansbasedonavailabletimeandexaminationschedules.
- NocentralizedplatformtosecurelystoreAI-generatedlearningresources.

● Difficultyassessingknowledgethroughself-generatedquizzes.

● Switching between multiple applications for note-taking, quizzes, and scheduling reduces productivity.

Solution

The AI StudyBuddy backend provides a centralized, AI-powered learning platform that automates several academic tasks through intelligent content generation. The system enables authenticated userstouploadorenterstudymaterialsandleveragetheGoogleGeminiAIAPItogenerateeducational resourcesautomatically.

● AuthenticatesusersusingJWT-basedsecurity.

● StoresuserinformationandstudyresourcesinMongoDB.

● SendsstudycontenttotheGoogleGeminiAImodelthroughdedicatedAIservicemodules.

● Producesmultiple-choicequizzesforself-assessment.

● Generatespersonalizedstudyplansbasedonuserpreferencesandexaminationtimelines.

● Storesgeneratedlearningresourcesforfutureretrievalandcontinuouslearning.

Usage

Student Usage

Students register and log in to the application before accessing AI-powered learning features. After authentication, they can upload studymaterial or provide textual content and choose the desiredAI functionality, such as generating summaries, quizzes, flashcards, or personalized study plans. The generatedcontentcanbereviewedimmediatelyandstoredforfuturereference.

Administrator Usage

Administrators manage registered users, monitor application activity, oversee system usage, and maintainplatformintegrity.Administrative privilegesincludemanaging useraccounts,monitoringAI serviceutilization,reviewingsystemlogs,andensuringsmoothbackendoperations.

Outcome

Byintegrating artificial intelligence with a secure backend architecture,AI StudyBuddy significantly improves the learning experience for students. Manual preparation time is reduced through automated content generation, allowing students to focus more on understanding concepts rather thanorganizingstudymaterials.

As aresult, AI StudyBuddy serves as ascalable andintelligent educational platform thatenhances productivity, promotes effective revision, and supports personalized learning for students preparing foracademicexaminationsandcompetitiveassessments.

Software Requirements

● OperatingSystem:Windows10/11,macOS,orLinux(supportscross-platformoperations).

● Node.js(v16orabove):Runtimeecosystemrunningserverlogicandroutinginfrastructure.

● npm(v8orabove):Nodepackagemanagertocontroloperationaldependencies.

● Express.js:LightweightroutingwebframeworktoconstructbackendRESTfulentrypoints.

● MongoDB:DocumentNoSQLenginestoringusers,posts,categories,comments,andanalytics metrics.

● Postman:APIverificationtoolkittoassertschemavalidationresponsesacrossadministrative routeshields.

● CodeEditor:VisualStudioCodeorsimilarIDEenvironment.

Hardware Requirements

● Processor:IntelCorei5(8thGenorabove)/AMDRyzen5orequivalent.

● RAM:8GBminimum(16GBrecommendedforconcurrentinstancesofMongoDB,processing layers,andexecutionnodes).

● Storage:1GBofavailablediskworkspace.

Epic 1: Project Architecture

The AI StudyBuddy backend follows a modular RESTful architecture designed to provide secure, scalable,andAI-powered learning services.Thesystemprocessesclientrequeststhrough Express.js, validatesusersusingJWTauthentication,interactswiththeMongoDBdatabasefordatastorage,and communicateswiththeGoogleGeminiAIAPItogenerateintelligentstudyresources.

ComponentsDefinition

• ClientApplication:Thefrontend(React.js)allowsusersto register,login, uploadstudy material,and accessAI-powered featuressuch as summaries, quizzes,flashcards,and study plansthrough REST APIs.

• Express Server: The Express.js server receives HTTP requests, handles routing, processes middleware,andsendsappropriateresponsestotheclient.

• AuthenticationMiddleware:JWT-basedmiddlewareverifiesuseridentity,protectsprivateroutes,and ensuresthatonlyauthenticateduserscanaccessAIservicesandstudyresources.

• RouteHandlers:Routes manage API endpoints related to user authentication, study materials, AI features,andadministrativeoperations.

• AIService Layer: A dedicated AI service communicates with the Google Gemini API to generate summaries,quizzes,flashcards,andpersonalizedstudyplansbasedontheuser'sstudycontent.

• DatabaseLayer:MongoDB,togetherwith MongooseODM,storesuserinformation,uploadedstudy materials,andAI-generatedlearningresourcessecurely. Entity-Relationship (ER) Diagram Description TheAIStudyBuddy backendmanagesdatathroughinterconnectedMongoDBcollectionsthatstore user information, study materials, and AI-generated learning resources. Each entity is linked using logicalrelationshipstoensureefficientdatamanagementandretrieval.

1.User Entity

Storesthedetailsofregisteredusersandadministrators.

Attributes:

- _id(PrimaryKey-ObjectId)
- name(String,Required)
- email(String,Required,Unique.
- password(String,Hashed)
- role(String-Student/Admin)
- createdAt

2. Study Material Entity

Storesstudycontentuploadedbystudentsfor AIprocessing.

Attributes:

- _id(PrimaryKey)
- userId(ForeignKey → User)
- title
- subject
- content
- createdAt

3. SummaryEntity

StoresAI-generatedsummaries.

Attributes:

- _id(PrimaryKey)
- userId(ForeignKey → User)
- materialId(ForeignKey → StudyMaterial)
- summary
- createdAt

4. Flashcard Entity

StoresAI-generatedflashcards.

Attributes:

- _id(PrimaryKey)
- userId(ForeignKey → User)
- materialId(ForeignKey → StudyMaterial)
- question
- answer
- createdAt

5. Quiz Entity

StoresAI-generatedquizzes.

Attributes:

- _id(PrimaryKey)
- userId(ForeignKey → User)
- materialId(ForeignKey → StudyMaterial)
- questions
- createdAt

6. Study Plan Entity

StorespersonalizedstudyplansgeneratedbyAI.

Attributes:

- _id(PrimaryKey)
- userId(ForeignKey → User)
- studyPlan
- examDate
- createdAt

Key Relationships

- StudyMaterial → Summary(One-to-One):Each uploaded material can have one AI-generated summary.
- StudyMaterial → Flashcards(One-to-Many):Onestudymaterialcangeneratemultipleflashcards.
- User → StudyPlan(One-to-Many):Ausercangeneratemultiplestudyplansbasedondifferent subjectsorexaminationschedules.
- User → AIResources(One-to-Many):AllAI-generatedsummaries,flashcards,quizzes,andstudy plansareassociatedwiththeuserwhogeneratedthem.

Key Features

- SecureUserAuthentication:ImplementsJWT-basedauthenticationwithencryptedpasswords usingbcryptforsecureuseraccess.
- StudyMaterialManagement:Allowsuserstouploadandmanagestudymaterialsthatserveas inputforAI-poweredlearningfeatures.
- AISummaryGeneration:Generatesconciseandeasy-to-understandsummariesfromuploaded studycontentusingtheGoogleGeminiAPI.
- AIFlashcard Generation: Automatically creates flashcards from study materials to improve conceptrevisionandactiverecall.
- AIQuizGeneration: Produces multiple-choice quizzes based on uploaded content, enabling studentstotesttheirknowledge.
- Role-Based Access Control (RBAC): Provides different access privileges for Students and Administratorstoensuresecuresystemmanagement.
- MongoDB Data Storage: Efficiently stores user information, uploaded materials, and AI-generatedlearningresourcesusingMongoDB.

Roles andResponsibilities

Student

Studentsaretheprimaryusersofthesystem.Theycanregister,loginsecurely,uploadstudymaterials, andutilize AI-powered featuressuchassummarygeneration,flashcardcreation,quizgeneration,and personalized study plans. Students can also view and manage their previously generated learning resources.

Administrator

Administrators managetheoverallsystemandensureitssmooth operation. Theymonitorregistered users,manageuseraccounts,overseeapplicationusage,maintainsystemintegrity,andensurethatAI powered services function correctly. Administrators also have permission to perform system-level managementtasksandresolveuser-relatedissues.

User Flow

The AI StudyBuddy application begins with the user registering or logging into the system. After successfulJWT-basedauthentication,theuseris redirected tothedashboard,wheretheycanupload study materials and access AI-powered features. Based on the uploaded content, the system generatessummaries,flashcards,quizzes,orpersonalizedstudyplansusing theGoogleGeminiAPI

MVCArchitecturePattern

TheAIStudyBuddy backendfollowstheModel-View-Controller(MVC)architecturetoorganizethe applicationintoseparatelayers,makingthecodemodular,maintainable,andscalable.

Model Layer(Data Layer)

The Model layer defines the MongoDB collections using Mongoose schemas. It is responsible for managing user information, uploaded study materials, and AI-generated learning resources while ensuringdatavalidationanddatabaseinteractions.

View Layer(API Response Layer)

Since AI StudyBuddy is abackend RESTAPI, there isno traditional userinterfacein the Viewlayer. Instead, the API returns structured JSONresponsesto theReact frontend,allowing users tointeract withtheapplicationthroughRESTendpoints.

Controller Layer (Business Logic)

TheController layer processes incoming API requests, validates user input, performs authentication and authorization, communicates with the database models, invokes the Google Gemini AI service whenrequired,andreturnstheappropriateresponsetotheclient.

MVCBenefits

TheMVCarchitecture separates data management,businesslogic,and clientcommunication into independent components. This improves code readability, simplifies debugging and testing, and allowsnewfeaturestobeintegratedwithoutaffectingexistingmodules.

Epic 2: ProjectSetup and Configuration

Creating the Project Folder

1. CreateanewfoldernamedAIStudyBuddy.
2. OpentheprojectfolderinVisualStudioCode.
3. InitializetheprojectusingNode.jsandinstalltherequireddependencies.

Project Initialization

OpentheprojectfolderintheVSCodeterminalandexecute: **npminit-y**

Installtherequireddependencies:npminstallexpressmongoosebcryptjsjsonwebtokencorsdotenvmulter @google/genai

InstallNodemonfordevelopment: npminstall--save-devnodemon

Create the Following Files

- index.js
- .env
- package.json
- .gitignore

EnvironmentConfiguration

Createa.envfileintheprojectrootandconfiguretherequiredenvironmentvariables. PORT=5000 MONGO_URI=your_mongodb_connection_string JWT_SECRET=your_jwt_secret GEMINI_API_KEY=your_gemini_api_key

Epic 3: Backend Development

The AI StudyBuddy backend is developed using Node.js, Express.js, and MongoDB, following a modular architecture. The application separates routing, controllers, middleware,models,andutility modulestoimprove codemaintainabilityandscalability.

CoreSystemScriptImplementations

1.ApplicationEntryPoint(index.js)

Theindex.jsfileservesasthemainentrypointoftheapplication.ItinitializestheExpress server,loadsenvironmentvariables,configuresmiddleware,establishestheMongoDB connection, registersAPI routes, andstartstheserverafterasuccessfuldatabase connection.

2.DatabaseConnection(src/utils/db.js)

ThismoduleestablishesaconnectionbetweentheapplicationandMongoDBusing Mongoose. It ensures that thebackendcommunicateswith thedatabasebefore processinganyclientrequests.

3.AuthenticationMiddleware(src/middleware/auth.js)

TheauthenticationmiddlewareverifiesJWTaccesstokensforprotectedroutes. It authenticatesusersbeforegrantingaccesstosecuredAPIsandincludesrole-based authorizationtorestrictadministrativeoperations.

4.AIIntegration(src/utils/gemini.js)

ThisutilityintegratesthebackendwiththeGoogleGemini2.5FlashAPI.Itsendspromptsto theAImodelandreturnsgeneratedresponsesforsummaries,flashcards,quizzes,and studyplans.

5.FileUploadMiddleware(src/middleware/upload.js)

Theuploadmiddlewarehandlesstudymaterialuploadsandvalidatesincomingfiles beforetheyareprocessedbythebackend.

BackendDevelopmentDemo: AI_Studybuddy_Backend_Demo.mkv

Epic 4: Database Configuration

The AI StudyBuddy backend uses MongoDB as its primary NoSQL database and Mongoose ODM for data modeling and schema validation. MongoDB stores user information, uploaded study materials, and AI-generated learning resources in a structured manner

InstallMongooseusingthefollowingcommand:npminstallmongoose

Create Database Connection

Beforeperforminganydatabaseoperations,establishaconnectionbetweenthebackend and MongoDB using Mongoose. The connection module is initialized during server startuptoensurereliablecommunicationwiththedatabase.

Mongoose SchemaDefinitions

1.**User Schema(models/User.js) :**The User schema stores authentication and profile information for registered users. It includes fields such as name, email, password, and role. Passwords are securely encryptedbeforebeingstoredinthedatabase.

2.Study Material Schema(models/Material.js): The StudyMaterial schema stores the learning content uploaded by users. This content acts as the input forAI-powered features such as summaries, quizzes, flashcards, and studyplans.

DatabaseCollections

TheAIStudyBuddydatabaseconsistsofthefollowingcollections:

● Users–Storesuseraccountandauthenticationdetails.

● StudyMaterials–Storesuploadedlearningmaterials.

APIRouteInterfaceEndpoints

ThebackendexposesRESTfulAPIstointeractwiththedatabaseandAIservices.

1.AuthenticationAPIs :Handlesuserregistration,login,JWTauthentication,andprotectedrouteaccess.
2.StudyMaterialAPIs : Allowsuserstoupload,retrieve,update,andmanagestudymaterials.

3.AIFeatureAPIs

Providesendpointsfor:

● AISummaryGeneration

● AIQuizGeneration

4. Administrative APIs: Provides administrator-only endpoints for managing users and monitoring system activities.

Epic 5: Project Execution

TheAIStudyBuddy backendcanbeexecutedlocallyusingNode.js.Before runningthe application,ensurethatMongoDBisrunningandtherequiredenvironmentvariablesare configuredcorrectly.ThebackendexposesRESTAPIsthatcanbetestedusingPostman orThunderClient.

Runningthe BackendServer

Installallrequireddependencies: npminstall

Startthedevelopmentserver: npmstart

After successful execution, the server connects to MongoDB and starts listening for incomingAPIrequests.

API Testing

ThebackendAPIscanbetestedusingPostman.

1.**User Registration** Registersanewuserbyacceptingtherequireddetailsandsecurelystoringtheencrypted passwordintheMongoDBdatabase.

Endpoint: POST/api/auth/register

2.**UserLogin**

AuthenticatestheregistereduserandgeneratesaJWTaccesstokenforaccessing protectedAPIs.

Endpoint: POST/api/auth/login

3 . Upload Study Material

Allows authenticated users to upload study material that will be used as input for AI-poweredcontentgeneration.

Endpoint: POST/api/material/upload

4 . Generate AI Summary

Sends the uploaded study material to the Google Gemini API and generates a concise summary.

Endpoint:POST/api/materials/:id/summarize

5.GenerateAIFlashcards

CreatesAI-generatedflashcardsfromtheuploadedstudymaterialforeffectiverevision.

Endpoint: POST/api/ai/flashcards

6.GenerateAIQuiz

Generates multiple-choice questions from the uploaded study material to help users assesstheirunderstanding.

Endpoint: POST/api/ai/quiz

7. Generate AI Study Plan

Createsa personalizedstudyschedulebasedontheuser'slearninggoalsandavailable studytime.

Endpoint: POST/api/ai/study-plan